Where the Headroom Lies: Block ℓ_p-Norm Statistics for Cyclic Join Bounds

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Abstract*—Pessimistic cardinality bounds are usually improved by strengthening the *program* combining degree statistics, from fractional edge covers to LpBound's Shannon-cone LP. We ask where the headroom lies for cyclic joins, using LpH-CEG, a fractional block-cover LP over ℓ_p-norm degree sequences of multi-way blocks, and find substantial headroom in block statistics. On single-relation statistics little is left: on a fixed hyperpath the cover LP is provably optimal among weightings valid on every self-join-free database, and LpBound's flow LP reaches the polymatroid optimum. Block statistics, degree sequences of two-relation joins conditioned on their interfaces, enter that program unchanged. With eleven norms on both sides they tighten LpBound by a median 236–10,200× on 120 rule-selected Hetionet cycles ($|V|$=6–10) and 9.7–87× on 120 IMDb cycles, below LpBound on every Hetionet and IMDb query, 15× on G-CARE's AIDS (241 queries) and 12× on YAGO (143 with agreeing counts). On CorrBound's 400 subgraph-matching queries they are a median 3.0× below its published estimates. A median 48% (Hetionet) and 79% (IMDb) of the log-gain lies outside a pairwise inner-product proxy for CorrBound's statistics, and all statistics but one avoid the join: with one sparse product per block the full catalog costs about 8× LpBound's collection, against 25–54× with materialized joins. We prove validity via an ℓ_p Composition (Shannon-flow) Inequality; an ℓ_p flow LP gives a polynomial-time candidate, checked per query. Limits: statistics come from data slices, guaranteed polymatroid bounds stop at $|V|$=12, and bound-guided join orders lose to DuckDB on IMDb.

***Index Terms*—cardinality estimation, cyclic queries, ℓ_p-norms, block statistics, polymatroid bound, pessimistic estimation**

I. Introduction

Cardinality estimates drive query optimization; their errors compound multiplicatively across joins, and cyclic queries can expose cross-relation correlations missed by independence assumptions.

Pessimistic estimation replaces estimates by guaranteed upper bounds [1]. The AGM bound [2] uses fractional edge covers; PANDA [3] and the degree-based MOLP bound [4] add max-degree (ℓ_∞) constraints; and most recently, **LpBound** [5] generalizes to all ℓ_p-norms of degree sequences, achieving orders-of-magnitude improvements on skewed data via an LP over entropy inequalities. Independently, statistics over small multi-relation joins [6], [7] capture joint constraints across relations, and *block extensions* (Section II-C) group several relations into one extension step.

The two lines improve the bound from different sides. LpBound strengthens the *program* over single-relation statistics; CEG$_M$, block extensions and CorrBound [8] enrich the *statistics* with multi-relation information, but only through max degrees or inner products of degree vectors. Between them sit the interface-conditioned ℓ_p norms of two-relation joins, which CorrBound's shared-interface products overlap only in part (Section VI). We ask which side has the headroom on cyclic joins, what those statistics add, and at what cost.

a) Our contribution: We separate what a better *program* and what richer *statistics* contribute to cyclic join bounds, and find the headroom on the statistics side. The instrument is LpH-CEG, a *fractional block cover LP* over ℓ_p-norm degree sequences of multi-way blocks: on cyclic queries an attribute is covered by several steps, and the slack admits fractional weights $w_j < 1$ that, with intermediate norms (ℓ_2), tighten the max-degree bound (w=1, p=∞). On single-relation statistics it cannot beat the entropic optimum (Figure 1); what it exposes is that *block statistics* move the bound far below that optimum, and that on them a stronger program pays off again. The estimator this leads to is block statistics with polymatroid inference.

Our contributions are:

1) **An estimator that moves the bound by orders of magnitude (Sections V-C, V-E).** With eleven norms on both sides, block statistics $\mu_p(F \mid I)$, $|F| \geq 2$, under the polymatroid program are a median 236–10,200× below LpBound on 120 rule-selected Hetionet cycles ($|V|$=6–10), 9.7–87× on 120 IMDb cycles, 15× on G-CARE's AIDS and 12× on the YAGO queries whose counts agree with G-CARE's, and a median 3.0× below CorrBound-e's published estimates on CorrBound's own 400 subgraph-matching queries, never below our exact counts. On Hetionet and IMDb block statistics a stronger program pays off again (5.0–11.6× over the cover LP, against 2.7–6.8× on single-relation statistics).
2) **Which statistics matter, and what they cost (Section V-C).** All interfaces of a pair block but one come from a scan of one relation with degree lookups in the other; the common-neighbour sequence needs one sparse product, and its ℓ_2 is determined by none of the others (extended version). A median 48% (Hetionet) and 79% (IMDb) of the log-gain lies outside a pairwise proxy for CorrBound's statistics, and the full catalog costs about 8× LpBound's collection (25–54× with materialized

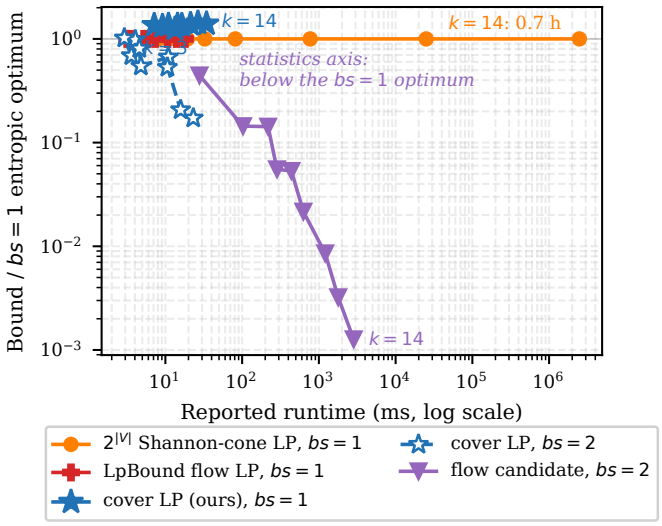


Fig. 1. Bound tightness and recorded runtimes (synthetic cycle-k queries; y relative to the entropic optimum on single-relation statistics). Timings have different scopes: bs=1 cover and Shannon-cone runs include collection; flow and bs=2 cover runs exclude it. The chain max-degree baseline is 2.3–971× looser than the cover LP ($k\leq$10; not plotted: its time was not recorded); the cover LP is within 1.43×, and the optimum is reached by LpBound's flow LP in 3–18 ms. Below y=1, the statistics axis: the same programs on bs=2 block statistics (three norms), the flow LP 2–800× below.

pair joins).

3) **Why single-relation statistics leave little headroom (Sections IV, IV-A).** An ℓ_p Composition Inequality makes the cover bound valid for every hyperpath, block size and p; on a fixed hyperpath the cover LP is optimal among weightings valid on every self-join-free database (Proposition 3), yet sits 2.7–6.8× above the entropic optimum, which LpBound's flow LP reaches in polynomial time. Our ℓ_p flow LP is a polynomial-time candidate for the polymatroid optimum over block statistics (equal on 633 of 862 queries), checked per query.
4) **Limits (Sections V, V-F).** Bound-guided join orders trim the catastrophic tail on small synthetic cyclic queries (P95 slowdown 1.2× vs. 5.8×, from the denser shapes) but lose to DuckDB's plans on IMDb; a per-step residual ordering heuristic does not generalize; and a predicate-free catalog helps only under mild predicates.

II. Preliminaries

A. Query Hypergraphs

A conjunctive query $\mathcal{Q}$ over relations $R_1, \ldots, R_m$ with attributes $V = \{a_1, \ldots, a_n\}$ is represented as a hypergraph $H = (V, \mathcal{E})$ where each hyperedge $e_i \in \mathcal{E}$ corresponds to the attribute set of relation R_i. The full natural join is $\mathcal{Q} = \bowtie_{i=1}^{m} R_i$, under set semantics (relations are sets of tuples).

B. CEG and MOLP

Cardinality Estimation Graphs [6], [7] model both optimistic and pessimistic estimators as path-finding in a directed graph. The pessimistic CEG$_M$ uses max-degree edge weights: for relation R with interface I, $\mu(R \mid I) = \max_{a_I} |\text{Ext}_R(a_I)|$. The MOLP bound [4] equals the minimum-weight path in the full attribute-subset CEG$_M$ [6]. Our experimental CEG$_M$ label denotes its restriction to whole-relation steps with nonempty interfaces after the first step (the *chain* baseline); it can be looser than MOLP even for binary queries (extended version).

C. Block Extensions and Hyperpaths

A *block* $F \subseteq \mathcal{E}$ is a connected subset of relations. The *block extension factor* is $\mu(F \mid I) = \max_{a_I} |\text{Ext}_F(a_I)|$, where $I = \text{attr}(S) \cap \text{attr}(F)$ is the interface with the current state. A *hyperpath* $P = (S_0{=}\emptyset, F_1, S_1, \ldots, F_m, S_m{=}\mathcal{E})$ decomposes the query into sequential block extensions. The product $\prod_{j=1}^{m} \mu(F_j \mid I_j)$ is a valid cardinality upper bound (the unit-weight ℓ_∞ case of Theorem 1, proved here). Our framework generalizes this product by replacing ℓ_∞ norms with LP-optimized ℓ_p-norms.

D. LpBound: ℓ_p-Norm Degree Sequences

LpBound [5] uses ℓ_p-norms of degree sequences as tighter statistics. For relation R with degree sequence $\mathbf{d} = (d_1, \ldots, d_k)$ where $d_i = |\text{Ext}_R(a_i)|$, the ℓ_p-norm is $\|\mathbf{d}\|_p = (\sum_i d_i^p)^{1/p}$. The core inequality is: for any $p \geq 1$,

$$\tfrac{1}{p}\, h(X) + h(Y \mid X) \leq \log \|\mathbf{d}\|_p$$

for the degree sequence $\mathbf{d}$ of Y given X in R. Optimizing over all valid inequalities via an LP yields bounds orders of magnitude tighter than a product of maximum degrees.

III. LpH-CEG Framework

A. ℓ_p-Norm Block Extension Factor

Definition 1 (ℓ_p Block Extension Factor)**.** *For a block F with interface I, let $\mathbf{d} = (d_1, \ldots, d_k)$ be the degree sequence where $d_i = |Ext_F(a_i)|$ for each distinct interface value $a_i \in \pi_I(\mathcal{Q}_F)$. The ℓ_p block extension factor is:*

$$\mu_p(F \mid I) = \|\mathbf{d}\|_p = \Big(\sum_{i=1}^{k} d_i^p\Big)^{1/p}$$

Special cases: $\mu_\infty(F \mid I) = \max_i d_i = \mu(F \mid I)$ (max-degree); $\mu_1(F \mid I) = \sum_i d_i = |\mathcal{Q}_F|$ (cardinality).

B. Fractional Block Cover LP

Given a hyperpath P with m steps, we define a variable $w_{j,p} \geq 0$ for each step j and each p-value. The LP minimizes the log-space bound:

$$\min \sum_{j=1}^{m} \sum_{p \in \mathcal{P}} w_{j,p} \cdot \log_2 \mu_p(F_j \mid I_j) \qquad (1)$$

subject to the fractional block cover constraints: for each attribute $a \in V$,

$$\sum_{p \in \mathcal{P}} \left(\sum_{j:\, a \in \text{attr}(F_j) \setminus I_j} w_{j,p} + \sum_{j:\, a \in I_j} \frac{w_{j,p}}{p} \right) \geq 1 \qquad (2)$$

Algorithm 1 LpH-CEG Estimation

Require: Query hypergraph H, relation data D, block size limit b, p-values $\mathcal{P} = \{p_1, \ldots, p_r\}$
Ensure: Upper bound $\hat{C}$ on $|\mathcal{Q}(D)|$

```
Enumerate blocks B ← {F ⊆ E : |F| ≤ b, F connected}
for each block F ∈ B and interface I do
    Collect d_{F,I} {binary pairs: no join tuples}
    Compute μ_p(F | I) for each p ∈ P
end for
if |V| ≤ 12 and the polymatroid LP solves successfully then
    return 2^{h*}, h* = max h(V) over the Shannon cone under all ℓ_p block constraints
end if
Build block-extension DAG with ℓ_∞ weights
Find candidate hyperpaths P_1, ..., P_K (shortest, then a bounded search) {K=1: default}
for each hyperpath P_i = (F_1, ..., F_m) do
    Solve LP (1)–(2) for weights (w*_{j,p})
    Ĉ_i ← ∏_{j,p} μ_p(F_j | I_j)^{w*_{j,p}}
end for
return min(Ĉ_1, ..., Ĉ_K, max-deg bound)
```

C. Why Cyclic Queries Benefit

a) Where the slack comes from: With binary extensions every attribute is new at exactly one step. On a star each leaf is new at one step and never re-covered, so its cover forces $w_j{=}1$ and the LP returns the max-degree bound. On a path an interior attribute is re-covered by the next interface, so the LP may trade part of an extension for that interface: at $p{=}1$ an AGM-style bound (on a path of three relations, weights $(1, 0, 1)$, i.e. $N_1 N_3$), and on skewed data intermediate norms too (on such a path with one heavy value, adding ℓ_2 lowers the bound from 1,000 to 330; extended version). On a cycle the closing step re-covers attributes already bound, so every attribute is covered by two steps; this is the source of Theorem 4, a sufficient condition for the gain, not a necessary one.

D. Algorithm

Algorithm 1 summarizes the LpH-CEG estimation pipeline.

The cover LP has $O(m|\mathcal{P}|)$ variables and $|V|$ constraints ($m \leq |\mathcal{E}|$; three norms by default), solved by HiGHS [9] in about 1–2 ms. Headline comparisons use eleven norms; CorrBound's workload uses 31. The polymatroid branch returns the estimator of Section V-C. The API `guaranteed_block_bound` takes precomputed constraints and a cover fallback; `estimate_lph_ceg` supplies the cover branch. Beyond $|V|{=}12$, or on solver failure, the latter provides the guarantee. Flow is an optional diagnostic, never the returned bound.

IV. Theoretical Results

Proofs not given in full here are in the extended version shipped with the artifact (code, data pipeline, results: `github.com/suanlab/LpH-CEG`).

Theorem 1 (ℓ_p Composition Inequality)**.** *Let $P = (S_0{=}\emptyset, \ldots, S_m{=}\mathcal{E})$ be a hyperpath, $\mathcal{P} \subseteq [1, \infty]$ (with $1/\infty{=}0$), and let $(w_{j,p})$ satisfy the cover constraint* (2)*, with the statistics computed exactly on the database D. Then*

$$|\mathcal{Q}| \leq \prod_{j=1}^{m} \prod_{p \in \mathcal{P}} \mu_p(F_j \mid I_j)^{\,w_{j,p}}$$

Proof. The empty-query case is trivial; otherwise let h be the entropy of the uniform distribution on $\mathcal{Q}(D)$, so $h(V) = \log_2 |\mathcal{Q}|$. Write $A_j = \text{attr}(F_j)$, $V_j = \text{attr}(S_j)$ and $E_j = A_j \setminus I_j$: the E_j partition V and $I_j \subseteq V_{j-1}$. Since $\pi_{A_j} \mathcal{Q} \subseteq \bowtie_{R \in F_j} R$, the ℓ_p inequality of [10] gives $\frac{1}{p} h(I_j) + h(E_j \mid I_j) \leq \log_2 \mu_p(F_j \mid I_j)$. Weight by $w_{j,p}$ and sum; with $w_j = \sum_p w_{j,p}$ and $s_j = \sum_p w_{j,p}/p$ it remains to show $\sum_j [s_j h(I_j) + w_j h(E_j \mid I_j)] \geq h(V)$, where (2) reads $w_{j(a)} + \sum_{j' : a \in I_{j'}} s_{j'} \geq 1$ for the step $j(a)$ that binds a. Let $g_j = h(E_j \mid V_{j-1})$ and $\delta_j = \max(0, 1 - w_j)$. (i) By the chain rule and submodularity, $h(I_{j'}) \geq \sum_j h(I_{j'} \cap E_j \mid V_{j-1})$. (ii) If $\delta_j > 0$, (2) makes $\{I_{j'} \cap E_j\}$, weighted $s_{j'}/\delta_j$, a fractional cover of E_j; $S \mapsto h(S \mid V_{j-1})$ is monotone submodular, hence fractionally subadditive, so $\sum_{j'} s_{j'} h(I_{j'} \cap E_j \mid V_{j-1}) \geq \delta_j g_j$. (iii) As $I_j \subseteq V_{j-1}$, $w_j h(E_j \mid I_j) \geq (1 - \delta_j) g_j$. Summing gives at least $\sum_j g_j = h(V)$. □

Only monotonicity and submodularity are used, so the argument holds over the whole polymatroid cone: the inequality is a Shannon-flow inequality [3], here proved constructively for this family. The redundant coverage that lets per-step weights fall below 1 (the source of every cyclic improvement, and the reason the elementary per-step argument ($w_j \geq 1$) does not apply) enters exactly as the fractional cover in (ii). The theorem assumes exact statistics of the database being bounded; the sketched ℓ_2 of the extended version is outside it.

a) Independent checks: Two checks that do not rely on the proof agree with it. Over 207,000 checks (every database with nonempty relations over a two-value domain for the triangle and 4-cycle, plus 15,000 random ones across six shapes, each at block sizes 1–3) the estimate never fell below the exact count (minimum ratio 1.0). And because the remaining inequality is linear in h and affine in w, an LP over the polymatroid cone decides it for every vertex of the feasible weight region: across 6,197 hyperpath structures on eleven shapes with $|V| \leq 6$ and block sizes 1–3 (1.73M vertices; every hyperpath on 19 of the 33 (shape, block size) pairs) the minimum slack is 0 to solver precision—tight, never negative. Reduced versions of both ship as regression tests.

Theorem 2 (Subsumption)**.** *LpH-CEG subsumes:*

1) *Max-degree block bounds: setting $p = \infty$, $w_j = 1$ recovers the product of block extension factors of Section II-C.*
2) *Binary ℓ_p-cover bounds: setting $|F_j| = 1$ gives LP* (1)–(2) *at bs=1—a polynomial-size relaxation of Lp-Bound's [5] entropic LP (Section IV-A), not necessarily its entropic optimum.*
3) *AGM bound [2]: with $|F_j|{=}1$ and $p{=}1$ on every step, each attribute's coverage is* 1 *per covering step, so* (2)

becomes the fractional edge-cover constraint and the objective becomes $\prod_j |R_j|^{w_j}$.

A. Relationship to LpBound

LpBound [5] also applies to cyclic queries and also solves a linear program over ℓ_p-norm degree statistics, so we must delineate precisely what the block-cover LP adds.

a) Two different LPs: LpBound's bound is the optimum of an *entropic* LP: it maximizes $h(V)$ over the Shannon cone (monotonicity and submodularity inequalities on all $2^{|V|}$ entropic terms) subject to the ℓ_p-norm statistics constraints [5], [10]. We follow LpBound in calling it the entropic bound, although the Shannon cone is the polymatroid relaxation of the entropic cone, so it is the strongest bound LpBound computes from those statistics, not necessarily the tightest they admit. Our LP (1)–(2) instead optimizes $O(m\,|\mathcal{P}|)$ weights over a *fixed hyperpath*, with one fractional-cover constraint per attribute. On identical statistics the entropic optimum is the tighter of the two by construction, and we do not claim otherwise: at $bs{=}1$ the cover bound is 1.25–1.43$\times$ looser on synthetic power-law cycles (Figure 2), 1.9–3.5$\times$ on six hand-picked real Hetionet cycles, and up to 8.8$\times$ on a cycle-8 repeating one relation, where the entropic cone exploits conditional structure the fixed hyperpath treats independently.

b) On a fixed hyperpath, reweighting is exhausted: Could a better weighting of the hyperpath's statistics close that gap? No: the cover constraint is also necessary.

Proposition 3 (Necessity)**.** *If nonnegative weights* $(w_{j,p})$ *make the product of Theorem 1 an upper bound on* $|\mathcal{Q}(D)|$ *for every database* D *whose atoms' relations are independent (no self-joins), they satisfy* (2)*. So on a fixed hyperpath the LP* (1)–(2) *is optimal over all weightings of its statistics that are valid on every database.*

Proof. Fix $a \in V$ and $N \geq 2$, and let D_N give a the domain $[N]$ and every other attribute a single value. Then $|\mathcal{Q}(D_N)| = N$, and $\mu_p(F_j \mid I_j)$ is $N^{1/p}$ if $a \in I_j$, N if $a \in A_j \setminus I_j$, and 1 otherwise. The bound is N^{c_a}, with c_a the left side of (2), and $N^{c_a} \geq N$ forces $c_a \geq 1$. □

(In entropy terms, D_N is the modular polymatroid $h(S) = [a \in S] \log_2 N$. With self-joins the copies' identity admits other valid weights, e.g. $|R|^3$ for a triangle on one relation R, which leaves an attribute uncovered.) A cutting-plane search with the polymatroid LP confirms this optimality on triangle, cycle-4/5/6 and K_4 (cycle-4: 15.335 in $\log_2$, against 14.858 for the polymatroid optimum). The residual gap comes from *statistics* the hyperpath leaves unused, which the polymatroid program exploits and Section V-C extends.

c) Efficiency: The Shannon-cone *formulation* of that optimum has $2^{|V|}$ variables and $O(2^{|V|}|V|^2)$ constraints and degrades accordingly: 81 ms at $|V|{=}8$, 25 s at $|V|{=}12$, 41 min at $|V|{=}14$, not run beyond (Figure 2; the same curve on real Hetionet degree sequences: 22 s and 27 min). But that is not what LpBound solves. Its LPflow [5, Thm. 4.5] rewrites every simple ℓ_p constraint (one conditioning attribute) as a flow bundle after Im et al. [11] and is polynomial; our reimplementation returns the entropic bound to 10^{-6} on every instance we compared (12 family–seed pairs, the sweep, the 240 long-suite queries) in 3–26 ms at $k{=}3$–16 (Figure 2). So at $bs{=}1$ the cover LP buys *no* asymptotic advantage over LpBound: as a $bs{=}1$ bound it is dominated (looser, and both solve in milliseconds). The fixed hyperpath buys the per-step decomposition (item 1 below) and the framework the statistics vocabulary (item 2), where inference becomes non-trivial again (Section V-E).

d) What the block-cover LP adds: What LpBound's single global program does not provide is structural:

1) **A per-step residual bound.** Because the LP is attached to a hyperpath, it yields a bound *per extension step*; resolved with a variable prefix projected out, it bounds the residual for every prefix value, the signal our WCOJ ordering consumes (Section V-F). LpBound could be resolved likewise but has no per-step decomposition.
2) **Block-level statistics.** With multi-way blocks ($|F_j| > 1$) the LP draws on ℓ_p-norms of *joined* block relations—statistics outside LpBound's single-relation vocabulary.

So at $bs{=}1$ LpH-CEG trades tightness for a decomposable bound (which our WCOJ ordering uses, without beating max-degree ordering in general; Section V-F); item 2 is the axis the paper is about (Section V-C).

Theorem 4 (Cyclic Improvement)**.** *Consider a cycle query* C_k *(*$k \geq 3$*) evaluated by binary extensions in cycle order, where relation* R_j *has cardinality* $N_j = |R_j|$*, maximum degree* M_j*, and* ℓ_p*-norms* $\mu_p(R_j)$ *of its degree sequence from* a_j *to* a_{j+1}*. Then:*

1) ***(Standard bound.)*** *The max-degree bound is*

$$B_{\max} = N_1 \prod_{j=2}^{k-1} M_j,$$

where the first extension has empty interface and contributes the cardinality N_1*, the* $k-2$ middle *extensions each contribute a max-degree* M_j*, and the cycle-closing extension (interface of size two, empty extension set) contributes a factor of* 1*.*

2) ***(LP bound.)*** *For every* $p \in \mathcal{P}$ *(with* $1 \in \mathcal{P}$*) the fractional block-cover LP* (1)–(2) *is feasible with weight* $p/(p{+}1)$ *and exponent* p *on every middle extension, together with the split* $(\frac{p}{p+1}, \frac{1}{p+1})$ *of unit weight across the first and closing* (ℓ_1) *extensions. This yields*

$$B_{\mathrm{LP}} \leq \min_{p \in \mathcal{P}} N_1^{\frac{p}{p+1}} N_k^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \mu_p(R_j)^{\frac{p}{p+1}},$$

e.g. $N_1^{2/3} N_k^{1/3} \prod_j \mu_2(R_j)^{2/3}$ *at* $p{=}2$*. (Within this family the exponent must be common, or non-increasing along the cycle: the cover constraint of the attribute between edges* j *and* $j{+}1$ *reads* $\frac{p_j}{p_j+1} + \frac{1}{p_{j+1}+1} \geq 1$*.)*

3) ***(Improvement.)*** *Therefore, for every $p \in \mathcal{P}$,*

$$\frac{B_{\max}}{B_{\mathrm{LP}}} \;\geq\; \Big(\frac{N_1}{N_k}\Big)^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \frac{M_j}{\mu_p(R_j)^{\,p/(p+1)}}.$$

At $p{=}2$ a middle factor exceeds 1 iff $\mu_2(R_j)^2 < M_j^3$: the sum of squared degrees is below the cube of the maximum degree, as when a few high-degree values carry the relation (n values of uniform degree d qualify iff $n < d$; many degree-1 values do not). At $p = \infty$, fractions involving p are interpreted by their limits; every factor tends to 1. The boundary factor depends only on the first and last relations; for a uniform *cycle ($N_j \equiv N$) it equals 1, and in the general case it is ≥ 1 for a hyperpath entering the cycle at the largest relation ($N_1 \geq N_k$). If every middle factor is at least $c > 1$, the ratio is at least $c^{\,k-2}(N_1/N_k)^{1/(p+1)}$: it grows geometrically in the number of middle edges. Since the cover branch returns $\min(B_{\max}, B_{\mathrm{LP}})$, the realized ratio is always ≥ 1.*

Proof sketch. The standard bound (1) follows directly from the cycle hyperpath structure: the first extension has empty interface, so its degree sequence is the single value N_1 and $\mu_\infty = N_1$; each middle extension adds one attribute over a size-one interface with $\mu_\infty = M_j$; the closing extension has both endpoints already in the state, so its conditional degrees are all 1 and $\mu_\infty = 1$. For (2), each attribute is covered as an extension in one step (rate 1) and as an interface in the next (rate $1/p$); with a common p the local constraint $w + w/p \geq 1$ is tight at $w = p/(p{+}1)$. Attribute a_2 then reads $w_1 + \frac{1}{p+1} \geq 1$, a_k reads $\frac{p}{p+1} + w_k \geq 1$ and a_1 reads $w_1 + w_k \geq 1$, all tight at $w_1 = \frac{p}{p+1}$, $w_k = \frac{1}{p+1}$. The closing step has $\mu_1 = N_k$, so the first/closing contributions are $N_1^{p/(p+1)} N_k^{1/(p+1)}$ against N_1 in $B_{\max}$. The LP optimum is at most every such point, and may mix p further. □

e) Observation: On power-law cycles with fixed parameters the middle factors exceed 1, so the bound predicts a ratio geometric in k; the measured one grows accordingly (10-seed means against the chain): 8.4× (k=4), 69× (k=6), 614× (k=8). Full MOLP grows far more slowly (1.4–5.1× above the cover LP, about 1.35× per two added vertices within each parity).

B. Tightness

Theorem 1 establishes that LpH-CEG bounds are *valid* (never underestimate), but not necessarily *tight* (matching a worst-case instance). Tightness depends on the degree sequence structure:

For finite p, each step's inequality is attained by $P(u) \propto d_u^p$ with a uniform conditional; for $p = \infty$, concentrate on a maximum-degree interface value.

Per-step tightness does not make the *composition* tight: the weighted product can be loose even where every factor is exact, and we measure it to be 1.25–1.43× looser than the entropic optimum at bs=1 on synthetic cycles and a median 2.7–6.8× on the real long suites, 59× on G-CARE's AIDS (Section V-C).

Block size is a trade-off: binary extensions keep simple interfaces; multi-way blocks capture block-level correlation at the cost of non-simple ones. Results use bs=1 unless marked bs=2.

C. Delta over Max-Degree Block Bounds

With ℓ_∞ weights and no optimization layer, the block extensions and hyperpaths of Section II-C give the max-degree block bound. What this paper adds is the ℓ_p fractional cover LP that replaces those fixed scalars, the separation of the statistics and inference axes (Section V-C), the flow-LP inference over block statistics (Section V-E); none exists in the max-degree bound. Theorem 4 additionally supplies a structural test that neither it nor LpBound provides, lower-bounding the magnitude by a product of $k-2$ per-edge factors $M_j/\mu_p(R_j)^{p/(p+1)}$ at a common p.

V. Experimental Evaluation

A. Setup

All experiments run on one server (Intel Xeon Gold 5218R, 20 cores, 251 GiB RAM; SciPy 1.15 with HiGHS, DuckDB 1.5.4); worker counts are stated where runs were concurrent. Bounds are floating-point LP optima, not exact-arithmetic certificates.

a) Datasets: We evaluate on six real-world datasets: **Hetionet** [12] (selected biomedical relations; 13 metaedges, 20K nodes, 438K edges); **Epinions** (social network, 75K nodes, 509K edges); **IMDb**, JOB's snapshot [13]: the 200 most-voted feature films for a cycle suite, and the JOB-medium subset; and the AIDS and YAGO graphs of G-CARE [14] with its cyclic queries; and DBLP through CorrBound's public workload (below). We also use synthetic power-law graphs (40–5,000 nodes, 120–20,000 edges). Seeds and runs are stated per table: multi-seed results are 10-seed means; single-instance tables say so.

b) Queries: Synthetic cycle-k for $k \in \{3, \ldots, 16\}$ plus diamond and K_4 families; on real data, three rule-selected, hash-pinned suites (Section V-D) plus a handful of hand-picked cycles.

c) Baselines: AGM (fractional edge cover, $\prod |R_i|^{w_i}$); CEG$_M$ (the whole-relation max-degree chain at bs=1, a restriction of MOLP [4], [6]); LpH-CEG (LP-optimized ℓ_p-norms, $\mathcal{P} = \{1, 2, \infty\}$). Every program in a comparison gets the same $\mathcal{P}$, including our entropic LpBound. Because the entropic bound gains more from extra norms than the cover LP does, the comparisons against LpBound (the abstract, Table III's upper block, the short suite, Tables II, IV and V, the pairwise CorrBound proxy, predicates, IMDb ordering) use the eleven norms $\mathcal{P} = \{1, \ldots, 10, \infty\}$ on both sides (CorrBound's workload: its 31 norms); every other comparison with the entropic bound (the other synthetic tables and figures, the six hand-picked cycles, the pair-size comparison, the 370-query suite) uses three norms, $\{1, 2, \infty\}$. The "LpH-CEG" column throughout denotes our *cover*-LP at bs=1 (Section IV-A); it is

TABLE I

LpH-CEG vs. the chain max-degree baseline on synthetic cycle queries (bs=1, 200V/800E, mean ± std over 10 seeds). The ratio is a *bound-tightening* measure (chain/LpH-CEG); LpH-CEG's q-error (accuracy vs. the true count) is shown only where we materialized the exact count, and is "—" for $k \geq 6$.

Query	Chain	LpH-CEG	Ratio	q-error
Cycle-3	53K±2K	19K±0.3K	2.75±0.13×	10.8±0.3×
Cycle-4	3.8M±253K	448K±44K	8.43±0.78×	18.3±2.0×
Cycle-5	270M±18M	11.3M±308K	23.8±1.3×	39.0±1.3×
Cycle-6	18.8B±2.3B	272M±16M	68.9±6.4×	—
Cycle-7	1.4T±167B	6.5B±329M	211±20×	—
Cycle-8	97T±14T	158B±10B	**614±71**×	—

TABLE II

Statistics × inference, 40 vertices / 120 edges, seed 101 (12 family–seed pairs behave alike; see text). Both programs see the same catalog (every single relation, or every $bs \leq 2$ block and interface) with the same eleven norms. Every entry is a valid upper bound on the exact count.

	cover LP		polymatroid optimum	
Query	single	block	single	block
Triangle	1,083	668	806	**352**
Cycle-4	9,855	4,716	7,388	**1,355**
Diamond	9,769	4,281	7,388	**1,252**
K_4	9,125	3,376	6,320	**1,093**

not the entropic LpBound of Zhang et al. [5], against which we compare separately in Section IV-A (our cover bound is 1.25–1.43× looser on synthetic cycles, a median 2.7–6.8× on Hetionet and IMDb). FactorJoin and GNCE [15], [16] produce point estimates; COLOR [17] also includes max-degree bound variants, whose guarantees depend on whether its statistics are sampled; CorrBound, which enriches the entropic program with cross-relation statistics, is compared on its own workload and through a proxy (Section V-C).

d) Unified bound comparison: On one synthetic suite (table in the extended version) the four guaranteed-bound methods order as predicted: CEG_M (chain), AGM ≥ Cover-LP ≥ Entropic (Theorem 2), with AGM and CEG_M incomparable. Our cover-LP is 2.3–35.7× tighter than CEG_M on synthetic cycles, is also consistently tighter than AGM (e.g. 2.0× on cycle-6), and stays within 1.25–1.36× of the entropic (polymatroid) optimum on the same single-relation ℓ_p statistics on these cycle-3–6 instances (one instance per cycle).

B. Main Results: Cyclic Query Bounds

Table I shows the bs=1 cover LP against the chain: LpH-CEG achieves 2.8× to 614× tighter bounds than the chain on cyclic queries, but only 1.4–5.1× (2.6× at cycle-8) against full MOLP on the same instances, so most of the chain's looseness is its restriction rather than max-degree statistics (extended version); with tight standard deviations (relative std $< 12\%$) confirming stability across random graph instances. The improvement *scales with cycle size*: from 8.4× on cycle-4 to 69× on cycle-6 to 614× on cycle-8. It also *scales with data size*: cycle-4 improves from 5.9× (100V) to 34× (2KV), with the dependence on skew characterized in the extended version.

C. Statistics Versus Inference

The two axes can be separated directly: hold the program fixed and change the statistics, then hold the statistics fixed and change the program. Table II is that 2×2, over four families and three seeds, with the exact join size computed so every bound is checked rather than assumed.

Reading the table by column pairs gives the central observation. At fixed program, the statistics axis (single → block) improves the bound by 1.5–2.7× for the cover LP and 1.8–5.9× for the optimum; at fixed statistics, the inference axis (cover → optimum) improves it by 1.2–1.5× on single-relation statistics and 1.6–3.8× on block ones, over all 12 pairs (three norms give the same picture: 1.5–2.3, 1.7–5.5, 1.15–1.5, 1.5–3.8). The axes interact: richer statistics give the program more to exploit. The statistics axis has room because a block statistic $\mu_p(F \mid I)$ with $|F| \geq 2$ describes a multi-relation join, which no single-relation degree *sequence* determines (most of it follows from one relation's tuples and the other's degrees; see below).

Two scoping points keep this from being overstated, both measured on the headline cycle protocol (200V/800E, 10 seeds; ratios of seed means over a different seed set from Table I's, one of ten shared, so cycle-8 reads 654× here and 614× there; the inference row is Figure 1's 100V/400E sweep):

Axis	cycle-4	cycle-6	cycle-8
both: max-deg → cover LP	8.8×	72.7×	654×
inference: cover LP → entropic	1.25–1.43× (3 or 11 norms)		
statistics: bs=1 → bs=2	1.21×	1.87×	2.70×

First, the claim concerns headroom *remaining once an LP is in place*. The first step, from the chain's max-degree product to the cover LP, dwarfs the others (≤5.1× against full MOLP) but changes the norms as well as the program (with $\mathcal{P} = \{\infty\}$ the LP gains 1.00×). Second, the separation between the two remaining axes is regime-dependent: on these homogeneous synthetic cycles they are comparable (1.2–2.7× against 1.25–1.43×), and it is heterogeneous real data that can separate them: on the 109-query short suite the statistics step (cover bs=1 → bs=2) is 30× against an inference gap of 2.8× with eleven norms (56× against 2.6× with three).

The diagonal comparison is the practical one. The *cheap* LP on rich statistics (4,716) beats the *optimal* LP on poor statistics (7,388)—on 12/12 family–seed pairs, and on all 109 cycle-3/4 queries of the registered Hetionet suite, where the median ratio to the entropic bound is 0.094× (0.057× with three norms). That subset and its 3K-row file-order prefix (to keep pair joins small) were chosen after registration, and cycle-5 was not run at bs=2. All 109 bounds were checked against exact counts with no violation, but the counts are only 0–334 (median 30) and bs=2 is still a median 85× above them: the ordering moved, not the distance to the truth—Table III is the uncapped test. This does *not* make the cover LP a better program (at

equal statistics the optimum wins on $12/12$ pairs, as it must): the choice of statistics comes first, and the strongest cell is the optimum on block statistics, i.e. LpBound on materialized pair-join views, with a polynomial-time candidate in Section V-E.

Table III extends the comparison to $|V|$=6–10 on two hash-pinned suites (Section V-D). With eleven norms on both sides the two axes diverge with $|V|$ on Hetionet: the inference gap stays at 2.7–3.1×, while bs=2 statistics are 46/270/920× tighter than the entropic bound at the median under the cover LP and 236/1,720/10,200× under the polymatroid program, on $120/120$ queries. With exact counts of 10^4–10^9, this is not a small-count artifact. Nor is it a slice artifact: on ten random 20K-row samples of every file the polymatroid medians are 233–244, 1,702–1,811 and 10,040–10,930×, and on six other IMDb film sets (lower popularity ranks, random draws) 26.7–32.7×, every query below LpBound and none below an exact count (extended version). Query-level bootstrap 95% intervals (over the registered queries of these fixed databases; 10,000 resamples: sensitivity to the query sample, not variability over databases) keep every median ratio to LpBound in this table and in Table IV on one side of 1, except the cover LP on block statistics on AIDS ($[0.42, 1.00]$); for the polymatroid program on block statistics that side is automatic (its constraints contain LpBound's), and the intervals of its gains are in the extended version. None of the 864 bounds with an exact count fails (216 checked queries over both suites × four bounds), nor the 345 three-norm ones (115 Hetionet queries × three). With three norms the entropic baseline is a median 1.9–3.2× looser and the margins correspondingly larger (80/630/2,500×, lower block). Two more comparisons (three norms, not tabulated) locate *which* statistic carries the gain. Keeping only the *sizes* of the same pair joins, $|R \bowtie S| = \sum_b \deg_R(b) \deg_S(b)$—the two-relation statistic CorrBound [8] adds to the entropic program, here added exactly (the constraints stay simple, so the flow LP is the polymatroid optimum) though without CorrBound's sketches or multi-relation terms—already beats the entropic bound by a median 15/27/99× (AGM over the same sizes: 7–15×). Under the same program (the flow LP, within 1.73× of the optimum, so these are lower bounds) the interface-conditioned block statistics add a further 19/80/154× (median over all 120: 63×) on $120/120$ queries (never below the exact count on the 115 with one).

On Hetionet the price is on the other side of the ledger: bs=2 statistics take 25–57 s at the median (up to 366 s; 33 queries exceeded a 180 s budget, which only flags them: each finished and is kept in every ratio), against under 1 s for bs=1 and 0.5–2.2 s for the entropic LP at these sizes. The last row measures what a system would materialize once: a workload-level catalog keyed by underlying table, orientation and interface holds 676 entries from 91 block builds, costs 527 s in total against the 11,765 s the per-query script spent, and returns the identical bound on $120/120$ queries. It saturates by the 68th query, after which a three-norm bs=2 cover estimate costs 7–10 ms, 100–220× below the $2^{|V|}$ entropic program (not LpBound's millisecond flow LP) at $|V|$=8–10 and below it on $60/60$ of the second half. The collection overhead is paid once per database snapshot.

IMDb. The second suite is a different kind of graph (a star schema in which every relation joins movies to people, companies or keywords) and separates the axes only with a program that can use the statistics. With eleven norms the inference gap is 5.1–6.8×, and bs=2 statistics are only 1.3/2.9/6.8× tighter than the entropic bound under the cover LP (level with or below that gap, and above the entropic bound on 8 of the 40 queries at $|V|$=6), but 9.7/30/87× under the polymatroid program, on $120/120$. On these statistics it beats the cover LP by 7.6–11.6×, against 5.1–6.8× at bs=1: richer statistics reopen the program here too. Under it the bound moves from 23–1,300× the truth to 2.2–15× (101 checked queries, exact counts $3{\cdot}10^5$–$9{\cdot}10^7$ at the median; 19 of the 80 at $|V|$=8–10 timed out; these are the largest, so the ratios to the truth describe the smaller queries). With three norms the entropic baseline is 3.5–6.7× looser and the cover-LP margin reads 4.0/13/43×. Pair-join sizes barely help (three norms: median 1.00×; they help at all on $24/120$, by at most 1.35×): half the pairs share an entity and so join one relation with itself, with size $\sum_x \deg(x)^2$, that relation's ℓ_2^2, which LpBound already has; the movie-sharing pairs are genuine inner products but rarely bind. The interface-conditioned statistics still add 17–171×. (IMDb has no catalog run; the catalog and its 7–10 ms estimate are Hetionet's.) At five times the films (exploratory, bounds only) every tightness ratio grows and collection by materialized joins grows faster than the data (48 s median per query, up to 25 min and 16 GiB on a shared machine; extended version), by sparse products less so (9.3 s, 83 s, 0.9 GB; Section V-C).

G-CARE. As an external check we run every cyclic query of the G-CARE benchmark [14] on its AIDS and YAGO graphs (241 and 366; query sets fixed by G-CARE, size caps below ours), with labelled, directed edges as binary relations filtered by the query's vertex labels, homomorphism counts, and eleven norms on both sides (Table IV). Our exact counts equal G-CARE's on $241/241$ AIDS queries; on YAGO they differ on 49 of the 192 we evaluate, in both directions and under every semantics we tried (directed or not, injective or not; over all 364 cyclic YAGO queries with a count, 80 differ). A DuckDB-free reimplementation reproduces our count on all 49, so the gap lies outside G-CARE's stated semantics, which we follow, and is unresolved (extended version). We check validity against our counts (one YAGO bound, 26, lies below G-CARE's 33 but above our 17). 174 YAGO queries exceed the prototype (a relation over 300K rows or a pair join over 20M) and are reported, not run. Under the polymatroid program, block statistics are 15× (AIDS) and 24× (YAGO) tighter than LpBound and never looser; the YAGO median mixes 12× on the 143 queries whose counts agree with G-CARE's and 311× on the 49 that do not, and inclusion and gain vary by topology (extended version). Under the cover LP the inference gap dominates on AIDS: 59×, plausibly as bounded molecular degrees make high norms very informative to the entropic program, against a 1.35× statistics gain (block statistics beat LpBound on only $135/241$); on YAGO the two

TABLE III

TWO RULE-SELECTED, HASH-PINNED REAL SUITES (IMDB AFTER A DISCLOSED PILOT), 40 CYCLES PER LENGTH ADMITTED IN SHA-256 ORDER OF THE CANONICAL RELATION SEQUENCE. HETIONET: METAGRAPH CYCLES OVER DISTINCT RELATIONS, THE FIRST 20K ROWS OF EACH RELATION IN FILE ORDER. IMDB (JOB SNAPSHOT): THE 200 MOST-VOTED FEATURE FILMS (A DENSE, POPULARITY-BIASED SLICE), NINE CAST ROLES, TWO COMPANY TYPES AND KEYWORDS, ONE RELATION EACH. MEDIANS OVER QUERIES; "POLYMATROID" IS THE $2^{|V|}$ PROGRAM AND "FLOW" OUR FLOW LP, BOTH OVER THE $bs{\leq}2$ CATALOG. UPPER BLOCK: EVERY PROGRAM WITH ELEVEN NORMS ($\mathcal{P} = \{1, \ldots, 10, \infty\}$), THE COMPARISON AGAINST LPBOUND; LOWER BLOCK: THREE ($\{1, 2, \infty\}$, THE COVER LP'S DEFAULT). EXACT COUNTS BY DUCKDB (60 S BUDGET); QUERIES THAT TIME OUT ARE UNCHECKED, NOT DROPPED, SO THE "/EXACT" ROWS ARE MEDIANS OVER $40/40/35$ AND $40/33/28$ QUERIES. IMDB TIMES WERE TAKEN UNDER 10 CONCURRENT WORKERS. THE CATALOGED ROW AMORTIZES BLOCK BUILDS OVER THE WORKLOAD; THE $|V|{=}6$ QUERIES, RUN FIRST, PAY MOST OF THEM.

	Hetionet			IMDb		
$\lvert V\rvert$	6	8	10	6	8	10
Eleven norms on both sides:						
cover $bs{=}1$ / entropic	2.66×	3.10×	3.00×	5.14×	6.75×	5.96×
cover $bs{=}2$ / entropic	0.022×	0.0037×	0.0011×	0.79×	0.35×	0.15×
polymatroid $bs{=}2$ / entropic	0.0042×	0.00058×	0.000098×	0.10×	0.034×	0.012×
of 40 below entropic, cover; polym.	40; 40	40; 40	40; 40	32; 40	40; 40	40; 40
entropic / exact	$8.4{\cdot}10^3$	$3.6{\cdot}10^5$	$2.8{\cdot}10^7$	23	171	$1.3{\cdot}10^3$
polymatroid $bs{=}2$ / exact	47	277	$2.8{\cdot}10^3$	2.2	6.4	15
Three norms on both sides:						
chain / cover $bs{=}1$	1.8×	5.1×	8.9×	1.8×	1.9×	3.2×
full MOLP / cover $bs{=}1$	1.4×	2.1×	3.6×	1.8×	1.9×	3.2×
cover $bs{=}1$ / entropic	2.53×	3.24×	3.61×	3.75×	4.33×	4.14×
cover $bs{=}2$ / entropic	0.012×	0.0016×	0.00039×	0.25×	0.075×	0.023×
$bs{=}2$ below entropic	40/40	40/40	40/40	40/40	40/40	40/40
cover / flow, both $bs{=}2$	3.2×	4.2×	5.1×	4.2×	4.0×	3.3×
flow = polymatroid	39	35	30	28	27	30
checked (invalid)	40 (0)	40 (0)	35 (0)	40 (0)	33 (0)	28 (0)
seconds, $bs{=}1$/ent/$bs{=}2$	0.6/0.5/25	0.8/0.8/57	0.8/2.2/33	0.3/0.3/4.4	0.6/0.8/14	0.9/3.3/22
$bs{=}2$, cataloged	0.56	0.007	0.010	—	—	—

are comparable ($6.1\times$ against $8.9\times$). With three norms the AIDS order flips ($3.4\times$ against $21\times$). On both graphs the $bs{=}2$ program gap ($12\times$, $2.7\times$) is below the $bs{=}1$ one. G-CARE is thus a regime where the program matters as much as the statistics.

CorrBound's workload. CorrBound's artifact [8] ships the 400 labelled 8-vertex subgraph-matching queries on DBLP that it and LpBound evaluate, with its estimates. With its norms ($p{=}1, \ldots, 30, \infty$) our LpBound reproduces the artifact's on $400/400$ queries, and its own LP code reproduces the published CorrBound-e ones. Block statistics under the polymatroid program sit a median $1.99\times$ above the exact count, against $6.1\times$ for CorrBound-e and $64.6\times$ for LpBound, and are tighter than CorrBound-e on $382/400$ (median $3.0\times$, at most $1.51\times$ looser); no bound falls below a recount (the shipped truths are a median 0.6% lower); ours cost a median 25 s per query, CorrBound's statistics are per dataset. For $p{>}1$ that code's inequalities are stronger than its Lemma 3.1; with the Lemma's, CorrBound-e is a median $2.4\times$ looser, and its statistics added to ours tighten $28/400$ queries by at most 3.1% (extended version).

a) Which statistic needs the join: Of the eight interfaces of a block of binary relations $R(a, b), S(b, c)$, seven (five non-trivial) come from a scan of one relation with degree lookups in the other (the degree of a is $\sum_{b \in N_R(a)} \deg_S(b)$, that of b is $\deg_R(b) \deg_S(b)$); only $\{a, c\}$, the common-neighbour count $|N_R(a) \cap N_S(c)|$, needs the pair join, and only its adjacency: one sparse product $A_R A_S$, not the join tuples. Its ℓ_2 is determined neither by the other interfaces nor by shared-attribute inner products (extended version), while its ℓ_1 is the join size. In the headline setting (eleven norms, polymatroid), the pair sizes and the pairwise products $\sum_b \deg_R(b)^q \deg_S(b)^r$ of interfaces $\emptyset$, $\{b\}$, $\{a, b\}$, $\{b, c\}$, $\{a, b, c\}$ ($(q, r) \in \{(1, 1), (p, p), (1, p), (p, 1)\}$: an exact proxy for CorrBound's inner products, not its full vocabulary) gain a median $42\times$ (Hetionet) and $2.0\times$ (IMDb) over LpBound; the one-side and common-neighbour statistics add a further $34\times$

TABLE IV

G-CARE CYCLIC QUERIES, ELEVEN NORMS ON BOTH SIDES (MEDIANS). YAGO QUERIES WITH A RELATION OVER 300K ROWS (144) OR A PAIR JOIN OVER 20M (30) ARE NOT RUN. EXACT COUNTS BY DUCKDB; G-CARE'S OWN COUNTS AGREE ON $241/241$ AIDS AND $143/192$ YAGO QUERIES.

	AIDS	YAGO
cyclic queries / evaluated	241 / 241	366 / 192
cover $bs{=}1$ / entropic	59×	6.1×
cover $bs{=}2$ / entropic	0.74×	0.11×
polymatroid $bs{=}2$ / entropic	0.066×	0.041×
cover $bs{=}2$ below entropic	$135/241$	$177/192$
polymatroid below / equal	241 / 0	188 / 4
entropic / exact	513	2,139
polymatroid $bs{=}2$ / exact	66	67
invalid bounds	0	0

TABLE V

FIVE CATALOGS, ONE SETTING (ELEVEN NORMS, POLYMATROID; MEDIANS, 120 QUERIES PER SUITE; EACH CATALOG IN ITS OWN PROCESS, SHUFFLED, 8 WORKERS, ONE RUN; THREE THREAD-PINNED REPEATS GIVE 7.8–8.5× SPARSE AND 23–52× MATERIALIZED): COLD COLLECTION (DATA LOAD EXCLUDED) RELATIVE TO LPBOUND'S (0.89/0.52 S; MEDIAN OF PER-QUERY RATIOS), GAIN OVER LPBOUND, BOUND OVER EXACT COUNT (115 AND 101 QUERIES WITH ONE; NONE BELOW IT).

	Hetionet			IMDb		
Catalog	collect	gain	/exact	collect	gain	/exact
LpBound	1×	1×	$2.1{\cdot}10^5$	1×	1×	127
join-free	6.0×	608×	772	6.9×	18.4×	7.7
+ common-nbr ℓ_2	6.8×	1,297×	340	7.6×	20.9×	7.3
full, sparse	8.2×	1,724×	211	8.3×	29.4×	4.6
full, materialized	54×	1,724×	211	25×	29.4×	4.6

and $14\times$ (48% and 79% of the log-gain) on $120/120$ queries, and alone reach $1{,}674\times$ and $28.7\times$ (full: $1{,}724\times$, $29.4\times$). The join-free seven carry 88% and 86% of the log-gain and the common-neighbour ℓ_2 leaves $1.47\times$ and $1.39\times$. Left out of the full catalog one group at a time, the one-side interfaces cost a median $1.8\times$ and $3.8\times$, the common-neighbour norms $2.6\times$ and $1.6\times$, sizes nothing ($120/120$), and shared products nothing on 98 and 85 of 120 (extended version).

Cost against tightness. One sparse product per block gives every common-neighbour norm, so the full catalog costs $8.2\times/8.3\times$ LpBound's collection, against $54\times/25\times$ with the pair joins materialized, with the same bound on $240/240$ queries and a median 60/44 MB of extra peak memory (at least) against 626/262 MB (Table V). The polymatroid program takes a median 0.08–0.10 s for every catalog, LpBound's flow LP 0.012–0.014 s. At five times the IMDb films ($|V|{=}6$) the sparse path takes a median 9.3 s per query (at most 83 s, peak 0.9 GB) against 48 s, 25 min and 16 GiB materialized (sparse: eleven norms; materialized: three), $5.1\times$ its time at 200 films for $4.4\times$ the rows.

A post hoc spectral reading of the common-neighbour ℓ_2 is in the extended version. On 22 small random cycles, a search over databases with the same $bs{\le}2$ statistics puts at most $1.5\times$ of the gap on the program and at least $2.5\times$ on the statistics (cycle-4 medians).

D. Real Data Results

At $bs{=}1$ on real data the LP fires only where cyclic structure meets cross-relation degree asymmetry (Section V-G). The heterogeneous Hetionet cycle-4, four relation types with asymmetric degrees, improves $7.59\times$ (53M→7.0M, relations capped at 30K edges); the C-D-G and D-G-A Hetionet triangles and the replicated Epinions cycles on 2–5K-edge file-order prefixes yield exactly $1.00\times$ (at 20K edges, in the registered suite below, the Epinions cycle-4 improves $2.06\times$). Over six closed metapaths of length 3–5 built from Hetionet's thirteen edge types, LpH-CEG fires *selectively*: it tightens C-G-G ($1.95\times$), C-D-G-G and C-D-A-G-G ($7.59\times$)—exactly those whose hyperpath places the skewed gene–gene relation in a *middle* position—and returns $1.00\times$ on the rest. The effect is *structural*: D-G-G-A holds the same high-skew relation as C-D-G-G yet yields $1.00\times$, in our reading because it sits on the closing edge, where Theorem 4 has no middle factor.

Beyond cycles. On a diamond (also run as a chordal 4-cycle, the same shape) and a 4-clique over the same edges, LpH-CEG improves 1.9–$4.2\times$ over the chain max-degree baseline (largest on the 4-clique), so the effect is not confined to cycles.

A pre-registered suite. A handful of hand-chosen queries cannot support a distributional claim, so we fixed a suite *before* measuring: every type-consistent metagraph cycle of length 3–5 over distinct Hetionet relations (370 queries, admitted by rule), plus seven Epinions shapes. The registry is hash-pinned and the evaluation refuses to run against an edited copy, so the selection could not be changed unnoticed after measuring. All 377 were evaluated and *none* censored. Over the 370 Hetionet queries the cover/entropic gap has median $1.99\times$, P90 $3.55\times$, maximum $6.29\times$ (215 distinct values), meeting the median $\le 3\times$, P90 $\le 8\times$ bar registered in advance. Improvement over the chain max-degree baseline has median $1.47\times$ and reaches $13.1\times$, yet $112/370$ queries (30%) improve by exactly $1.00\times$—the selective behavior above, now over a population rather than a handful. The Epinions part is degenerate and we report it so: one homogeneous relation self-joined makes all seven shapes yield an identical cover/entropic *gap* of $3.28\times$—one distinct observation rather than seven. A second registry extends the rule to lengths 6, 8 and 10, where the 777/4,380/12,960 admissible cycles are too many to solve entropically, so the first 40 per length in SHA-256 order of the canonical relation sequence are admitted (an order blind to data and bounds), giving the 120 Hetionet queries of Table III. A third registry, fixed after a disclosed pilot, uses IMDb from the Join Order Benchmark [13]: the 200 most-voted feature films (197–37K rows per relation), with one relation for each of nine of the twelve cast roles (costume and production designers and guests omitted when the relations were defined, before any bound was computed), for distributors, for production companies, and for keywords. Each relation joins movies to an entity type of its own, so a person reached as an actor and as a director are different variables: cross-role joins are excluded by design. The schema is then a star—a cycle alternates movies with entities, each entity reached by one relation on both sides, and relations repeat (JOB joins `movie_companies` and `title` twice); of the 364/3,081/25,752 such cycles at length 6/8/10 the first 40 in hash order are admitted. The pilot used the Hetionet rule (one person type, distinct relations) on the first 3,500 feature films of the year 2005; $82/120$ of its joins were empty, so these provided no informative positive-cardinality validity check, and scope and typing were both changed before registering. The pilot favored block statistics more (median cover $bs{=}2$/entropic $9.5{\cdot}10^{-6}$ against 0.077 here); it, and a run at five times the films (Section V-C), are in the artifact.

E. Scalability

On power-law cycles (100V/400E, median of 7 runs) the LP adds under 4.7 ms to the max-degree bound (1.17–$1.34\times$

overhead, 8–24 ms in total for k=3–10) while the improvement over the chain grows exponentially with k, reaching 971× at cycle-10 (full MOLP, run only on Table I's cycle-3–8 instances, is 1.4–5.1× above the cover LP).

a) Polynomial inference on block statistics: The Shannon-cone formulation of the entropic bound has $2^{|V|}$ variables and, on the power-law cycles of Figure 2, rises from 81 ms at k=8 to 25 s at k=12 and 2,480 s at k=14 (real Hetionet sequences: 22 s and 27 min), timed end to end with a sparse constraint matrix and HiGHS's interior-point solver. LpBound does not solve that program. Its LPflow [5, Thm. 4.5] follows Im et al. [11]: each *simple* constraint ($|X| \leq 1$) becomes a capacitated edge $X \to Y$ in a flow network over attribute sets (an ℓ_p constraint adds $\emptyset \to X$ at capacity δ/p), and the bound is the cheapest capacity assignment that routes one unit of flow to every attribute—a polynomial LP. Our reimplementation returns the entropic bound to 10^{-6} on all 12 instances of Table II and at every $k \leq 14$ here, in 3–26 ms for k=3–16; the cover LP is 1.25–1.43× looser than a program that is itself polynomial; both solve in milliseconds.

The question that remains open is inference over *block* statistics, where a bs=2 block over three attributes contributes constraints conditioned on two of them ($|X|$=2)—the non-simple case, which Im et al. show to be as hard as the general polymatroid bound already at $|X| \leq 2$, $|Y| \leq 3$ (their Theorem 1.3(c)). Their §7 relaxation handles such a constraint along a fixed attribute order π by restricting it to the attributes of Y that follow all of X; we carry the ℓ_p term through unchanged. The relaxed constraint is implied by the original (monotonicity), and for ℓ_∞ constraints every order's flow bound is valid by their Theorem 1.5(a); for ℓ_p we have no proof and certify each bound instead (numerically, to the solver's tolerance, not in exact arithmetic). The $2^{|V|}$ polymatroid bound over the *unrelaxed* constraints is valid for every p (each ℓ_p constraint holds on the join's entropy, as in Theorem 1), so a flow bound at or above it is valid; a flow bound counts as a guarantee only with this certificate. The released `guaranteed_block_bound` therefore returns the polymatroid optimum up to $|V|$=12 and the cover LP on the same statistics beyond (Theorem 1; the sweep's k=16), keeping the flow value only as a candidate. Every real-data flow bound passes (eleven norms: 109/109 short, 240/240 long-suite, 433/433 G-CARE and 80/80 resampled slice queries, equal on 633, to 10^{-6}), as do the synthetic sweep's at k=12 and 14 (equal; k=16, where the $2^{|V|}$ program does not finish, is uncertified) and 40 random connected shapes ($|V| \leq 5$, all 2,046 order–instance pairs, also never below the exact join). Two facts make the relaxation usable. The $|X|$=2 constraints carry most of the bs=2 gain (dropping them loosens the polymatroid bound 1.5–2.9×), and an order keeps one exactly when both conditioning attributes precede its target—so on a cycle an *alternating* order (one color class, then the other) keeps every constraint centered on the second class; no single order keeps them all, so we solve the flow LP over the $4|V|$ walk and alternating orders and keep the best bound. Over the full bs=2 catalog (every block, every interface) it equals

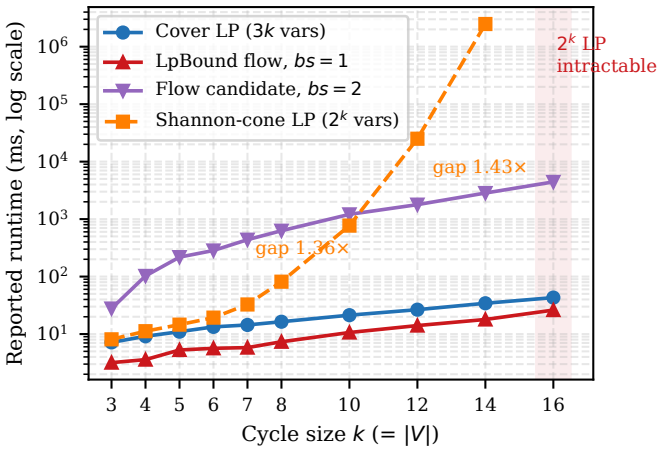


Fig. 2. Recorded runtime on cycle-k queries (log scale): bs=1 cover and Shannon-cone runs include collection, whereas flow runs exclude it. The Shannon-cone formulation of the bs=1 entropic bound grows rapidly in this measured range (25 s at k=12, 2,480 s at k=14), but LpBound's flow LP reaches the same bound (annotated gap to the cover LP) in 3–26 ms. On bs=2 block statistics the flow LP over $4k$ orders takes 0.03–4.4 s and, with three norms, equals the $2^{|V|}$ polymatroid bound at every even $k \leq 14$.

the $2^{|V|}$ polymatroid bound on 12/12 family–seed pairs (three norms; 7/12 with eleven) and on every synthetic even cycle up to k=14 (24 min for the $2^{|V|}$ program on this catalog at k=14) and on the triangle, and loses a constant 1.56–1.59× on cycle-5/7, where one wedge must remain lost; 34/40 of the random shapes are exact (max gap 1.28×). Cost is 4–73 ms per order and 0.03–4.4 s in total at k=3–16 (Figure 2). What it *finds* matters more than what it costs: the cover LP, confined to one hyperpath, leaves most of the block statistics unused, and the flow LP over the same catalog is 1.5–3.8× tighter on the small families (eleven norms: 1.6–3.8×) and, with three norms, 2.2× at k=3 growing to 288× at k=16 (uncertified) on the sweep (below the bs=1 optimum by 2–800× where the cover LP manages 1–6×; Figure 1). On real data the $2^{|V|}$ program over the same catalog is cheap enough to solve directly (eleven norms: median 0.02/0.18/2.4 s at $|V|$=6/8/10 on Hetionet, and 63 s for the slowest, a $|V|$=12 G-CARE query), so the eleven-norm real-data bounds we report are its optimum. The flow LP equals it on 633 of the 862 reported queries and is within 2.41× on all; beyond $|V| \approx 12$ it stays a candidate and the cover LP is the guarantee. On the 120-query Hetionet long suite the median gain of the polymatroid program over the cover LP on the same statistics is 5.0/6.7/11.1× at $|V|$=6/8/10; no bound falls below the exact count.

b) Choice of p-values: On the synthetic cycle-6 (100V/400E), $\mathcal{P} = \{1, \infty\}$ (2 values) already achieves 17.6× improvement; adding ℓ_2 ($\mathcal{P} = \{1, 2, \infty\}$) reaches 35.7×; further values (ℓ_3, ℓ_5) yield no additional gain. We recommend $\mathcal{P} = \{1, 2, \infty\}$ as the default (cardinality ℓ_1, second-moment information ℓ_2, worst case ℓ_∞); repeating the sweep on four families (cycle-4/6, K_4, chordal-5; extended version) confirms this is not cycle-specific—ℓ_2 helps in every family and p beyond $\{1, 2, \infty\}$ adds $< 2.4\%$ to the cover LP. This does not

TABLE VI
EXECUTION ROBUSTNESS (EXECUTION TIME ONLY): SLOWDOWN OVER THE BEST EVALUATED PLAN (DUCKDB, FORCED LEFT-DEEP ORDER VS. THE ENGINE'S OWN OPTIMIZER). LOWER IS BETTER; THE *tail* ROW IS THE WORST CASE OVER ALL QUERIES.

Query	Bound-guided	DuckDB opt.
Cycle-4 (100V/400E)	1.10×	0.60×
Cycle-5 (100V/400E)	1.52×	0.14×
K_4 clique (120V/800E)	1.64×	18.1×
K_4 clique (120V/1500E)	3.32×	21.0×
K_4 clique (120V/2500E)	1.81×	7.2×
K_4 clique (200V/2000E)	1.57×	18.3×
Diamond / chordal-4	2.54×	0.39×
Chordal 5-cycle	2.11×	0.13×
Hetionet △ (real)	3.56×	1.11×
Worst case (tail)	**3.6×**	**21.0×**

hold for the entropic bound, which gains a median 1.9–6.7× from eleven norms on the real suites; comparisons against it therefore use eleven norms on both sides (Section V).

F. Bound-Guided Join Ordering

Binary ordering in an engine. (Greedy WCOJ ordering by the per-step residuals is 1.6% worse at the median over 66 queries; extended version.) We load each query into DuckDB, compute a left-deep order by minimizing the classic C_{out} cost [18] (sum of the bound's estimated intermediate cardinalities), inject it by disabling DuckDB's join-order optimizer, and time real execution. As a reference we also run DuckDB's own optimizer (its cardinality estimates come from table statistics such as HyperLogLog distinct counts).

Two findings frame the result. A cardinality *upper bound* is a conservative cost model, so ranking plans by it need not match an engine optimizer on average (over all left-deep orders the cover LP also correlates less with true intermediate cost than the entropic bound: Kendall τ 0.27–0.87 against 0.33–0.94); and on a single cycle every proper sub-query is acyclic, where the cyclic slack of Theorem 4 is absent (intermediate norms can still help there, Section III-C), so that theorem's gain reaches the ranking only through *intermediate* sub-queries that are themselves cyclic (cliques, chordal cycles).

What the pessimistic bound *does* buy is robustness against catastrophic plans. Table VI reports, per query, each strategy's slowdown over the best plan we found. DuckDB's optimizer is excellent on average (often faster than any forced left-deep order because it builds bushy plans), but on dense 4-cliques its statistics-based estimates badly underestimate the cyclic intermediate blow-up, and it runs 7–21× slower than the best evaluated plan across every clique density we tried. Bound-guided ordering, whose estimates never fall below that blow-up, stays within 3.6× of the best evaluated plan ($\leq$25 sampled orders) on *every* query in the suite (DuckDB is faster on every non-clique row, the real triangle included). The worst-case (tail) slowdown is 3.6× for bound-guided versus 21× for the engine—a 5.9× heavier tail for the engine's optimizer.

a) The tail, over a broader distribution: Of six shapes at two sizes and three seeds, 24 queries survive (cycle-3/4, K_4, diamond; cycle-5 and chordal-5 censored entirely, by the AGM bound, which depends only on relation sizes, without using either strategy's estimates; this still restricts the shape and size distribution). Medians are close (1.03× vs. 0.93×) while the tails diverge: the bound-guided P95/max are 1.21×/1.31× against DuckDB's 5.82×/8.17× (with n=24, P95 is the third-largest value). The reference is the best *forced left-deep* order, so bound-guided is $\geq$1 by construction while DuckDB, free to go bushy, scores below 1 on median in this initial timing protocol: a bound-guided choice avoids a plan space's tail; it does not beat the engine. Five shuffled repetitions per plan keep the tail (P95/max 1.18/1.21× against 5.51/6.77×; per repetition P95 $\leq$1.65× against $\geq$5.07×; medians 1.03 and 1.18×), and so does a raw-logged replication that picks the reference on two runs and measures on four (P95 1.11 against 5.26×). The tail is shape-dependent: DuckDB's plan is faster than the bound-guided one on triangles and 4-cycles (median ratio 1.13, 1.78) and slower on diamonds and K_4 (0.53, 0.25; extended version).

b) Real data, block statistics: The same protocol on the 40 IMDb cycles at $|V|$=6 tests the thesis with the headline estimators behind C_{out} (eleven norms: LpBound's bound, the guaranteed $bs\leq$2 bound, exact counts), the chosen and 16 sampled left-deep orders timed five times each, one query at a time. The block bound picks the exact-count order on $19/40$ queries (the same order as LpBound's on 16), and on the 33 whose forced orders all finish within 60 s its median slowdown over the best timed order is 1.05×, against 1.08× for LpBound and 1.06× for exact counts (faster than LpBound's pick on 6, slower on 3, within 5% on 24). None competes with DuckDB's optimizer (0.26× median, at most 1.20×; on the other 7 nearly every forced order times out, so excluding them favors the forced orders), and the block statistics of every prefix take a median 12.5 s to collect against 2.2 s, far more than any execution they save. A better bound improves *which* left-deep plan is chosen, not whether one should be.

LpH-CEG is thus not a replacement for a well-tuned optimizer, but a guide that limits the worst case on the dense cyclic queries that defeat its estimates (on our synthetic cliques, not on IMDb), consistent with the pessimistic-estimation rationale of avoiding catastrophic plans [19].

G. When LpH-CEG Does Not Help

On our workloads LpH-CEG yields no improvement (1.00×) on (1) acyclic star queries (no redundant coverage) and (2) cycles of replicated relations with near-symmetric sampled degrees (the 5K-edge-prefix Epinions cycles, Section V-D, and all 8 IMDb co-actor/co-director and replicated movie–actor cycle-4 instances at 5K–1M rows); case (2) is a property of the sample, as 20K edges of the same graph give 2.06× on cycle-4 (Section V-D). On our data the gains came from *asymmetric degree structure in cyclic joins* (JOB-medium and TPC-H-like cases in the extended version). Synthetic power-law graphs improve more over the chain (up to 614× at cycle-8; 2.6× over full MOLP) than real data, where the median over the long suites is 1.8–8.9× (1.4–3.6× over

full MOLP) (tails $157\times$ on Hetionet, $34\times$ on IMDb): every synthetic relation is drawn with one shared Zipf exponent (e.g. $\alpha{=}1$), so every middle factor of Theorem 4 exceeds 1 at once, whereas real relations differ in skew.

VI. Related Work

a) Block decomposition methods: CEG's pessimistic variant CEG_M [6], [7] models estimation as a shortest path over max-degree weights of base relations and small joins (its optimistic CEG_O, like Graphflow's catalog [20], uses average degrees of such joins); the block extensions of Section II-C generalize this to groups of relations via hyperpaths. With max-degree weights, these pessimistic block bounds use ℓ_∞ statistics. LpH-CEG replaces these scalar weights with LP-optimized, interface-conditioned ℓ_p-norm products and gives a sufficient condition on the data for the LP to improve on cyclic queries with redundant coverage (Theorem 4).

b) Information-theoretic bounds: The AGM bound [2] uses fractional edge covers with relation cardinalities (ℓ_1-norms). PANDA [3] and MOLP [4] add max-degree constraints (ℓ_∞-norms). LpBound [5] generalizes to all ℓ_p-norms, showing that intermediate norms capture degree skew that max-degree misses. Since then, Im et al. [11] compute the polymatroid bound for simple degree constraints by a polynomial flow LP and show the non-simple case ($|X|{\le}2$, $|Y|{\le}3$, the shape of a $bs{=}2$ block constraint) as hard as the general polymatroid bound, which is why our flow bound over blocks fixes an attribute order; CorrBound [8] adds inner products of degree vectors across join columns (two-relation statistics in our terms, plus k-way products) to the entropic program and reports 6.7–$16.3\times$ lower error than LpBound on cyclic subgraph-matching queries; there, block statistics are a median $3.0\times$ tighter than its published estimates, and on our suites they add a median $34\times$ (Hetionet) and $14\times$ (IMDb) beyond an exact pairwise inner-product proxy (eleven norms; Section V-C). The ℓ_1-norm of a $bs{=}2$ block sequence *is* CorrBound's inner product (the join size). The shared-interface block norms $\sum_b \deg_R(b)^p \deg_S(b)^p$ are ℓ_p-weighted inner products. The one-side and common-neighbour statistics add information beyond the listed pairwise proxy: the extended version separates the common-neighbour ℓ_2 from those statistics, not from CorrBound's general Lemma 3.1. Its sparse product also has a different collection cost from CorrBound's sketches (Section V-C). Lin and Wang count bi-star [21] and caterpillar [22] homomorphisms into each relation: richer single-relation statistics, complementary to blocks (the extended version's separation fixes them too). SafeBound [23] deploys degree-sequence bounds [24] with predicates on relational workloads, following pessimistic CE [19]; we target graph patterns and a per-step decomposition.

Theorem 1 applies Abo Khamis et al.'s entropy framework [10] to block joins.

c) Summaries and learned estimators: COLOR [17] includes both average-degree estimates and max-degree bound variants; sampling can relax the latter's guarantees. GNCE [16], SPACE [25] and FactorJoin [15] learn point estimates, for which pessimistic bounds can provide a safety fallback [1]; PathCE [26] bounds over path summaries, which our programs do not take as input.

d) Join evaluation algorithms: Yannakakis+ [27] (acyclic queries) and CompactLTJ [28] evaluate joins; Capelli et al.'s sampling variant [29] takes tree-superadditive prefix bounds, which our residuals could supply if they qualify, though they do not generally beat max-degree ordering in our 66-query study.

VII. Practical Considerations

a) When to use LpH-CEG: Use it on cyclic queries with moderately skewed data, with $\mathcal{P} = \{1, 2, \infty\}$ for the cover LP (the entropic and polymatroid programs gain from more norms): gains are rare on triangles and grow with cycle length; without a cycle Theorem 4 does not apply, though intermediate norms can still help (Section III-C).

b) Under predicates: Block statistics of unfiltered relations stay valid under a selection (degrees only shrink) but loosen. With a production-year predicate on every movie variable of the 40 IMDb $|V|{=}6$ cycles, exact filtered single-relation statistics plus a predicate-free block catalog (eleven norms) put the polymatroid bound a median $1.9\times$ below LpBound on $40/40$ queries when 46% of rows pass (year$\ge$2005), but only $1.03\times$ $(22/40)$ at 18% (year$<$1995); per-predicate block statistics stay 4.2–$5.3\times$ below, and no bound falls below the exact count (240 checks). Selective predicates thus need per-predicate block statistics.

VIII. Conclusion

Added to LpBound's own program (eleven norms on both sides), block statistics tighten it by a median 236–$10{,}200\times$ on our Hetionet slice and 9.7–$87\times$ on IMDb, $15\times$ on G-CARE's AIDS and $12\times$ on the YAGO queries whose counts agree with G-CARE's, and are a median $3.0\times$ below CorrBound-e on its own workload; no bound fell below our exact counts. Single-relation statistics leave less: the cover LP is optimal among fixed-hyperpath weightings valid on every database without self-joins (Proposition 3) yet sits 1.25–$1.43\times$ (synthetic) and a median 2.7–$6.8\times$ on Hetionet and IMDb above the entropic optimum, which LpBound's flow LP reaches in polynomial time; richer statistics reopen the program. A median 48% and 79% of the log-gain remains after adding a pairwise inner-product proxy for CorrBound's statistics, and with one sparse product per block the full catalog costs about $8\times$ LpBound's collection. Our ℓ_p flow LP is a polynomial-time candidate (optimal on 633 of 862 queries), checked per query, not proved valid; past $|V|{\approx}12$ the cover LP is the guarantee.

Limits. Our evidence is binary graph patterns: Hetionet and IMDb slices ($\le$30K rows per Hetionet relation, 200–1,000 films), G-CARE's AIDS and YAGO (174 YAGO queries exceed our prototype) and CorrBound's DBLP queries. Our implementations support no general predicates (Section VII), so STATS-CEB [30] and JOB's queries are out of scope. Full-catalog collection grows faster than the data ($5.1\times$ the time for $4.4\times$ the rows). Bound-guided join orders lose to DuckDB on IMDb: this maps headroom; it is not an optimizer.

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Use of AI-generated content. In accordance with the IEEE policy on AI-generated content, we disclose that Claude (Anthropic), a large language model (Claude Opus models, used through the Claude Code tool), was used throughout this work. It drafted and revised text in every section (Sections I–VIII and the abstract); implemented and ran the experiment scripts behind Section V, all tables and figures; wrote the first versions of the proofs of Theorem 1 and Proposition 3 and of the extended version; wrote the automated audit that checks registered numerical claims against the result CSVs produced by the released scripts; and produced simulated reviews of drafts. The level of assistance was substantial: the model generated, and did not merely edit, much of the text, code and proofs, under the authors' direction. The authors directed the work, reviewed the content and take full responsibility for it. OpenAI Codex, another large language model, was also used in a later revision, at a lower level of assistance, to audit claims, references and code, recompute recorded results, and edit text in Sections I–VIII, the plotting code and the API checks.

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